

# class12

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Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes. Hint: The `read.table()`, `summary()` and `boxplot()` functions will likely be useful here. There is an example R script online to be used ONLY if you are struggling in vein. Note that you can find the median value from saving the output of the `boxplot()` function to an R object and examining this object. There is also the `median()` and `summary()` function that you can use to check your understanding.

The sample sizes for each genotype were: A/A: 108 samples A/G: 233 samples G/G: 121 samples

The median ORMDL3 expression values for each genotype were: A/A: 31.25 A/G: 25.06 G/G: 20.07

```
data <- read.table(  
  "rs8067378_ENSG00000172057.6.txt",  
  header = TRUE,  
  stringsAsFactors = FALSE  
)
```

```
# Sample size per genotype  
table(data$geno)
```

```
A/A A/G G/G  
108 233 121
```

```
# Median expression per genotype  
tapply(data$exp, data$geno, median)
```

```
      A/A      A/G      G/G  
31.24847 25.06486 20.07363
```

```
# (Optional) also show summary stats per genotype
by(data$exp, data$geno, summary)
```

```
data$geno: A/A
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
11.40  27.02   31.25   31.82  35.92   51.52
-----
data$geno: A/G
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 7.075  20.626  25.065  25.397  30.552  48.034
-----
data$geno: G/G
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 6.675  16.903  20.074  20.594  24.457  33.956
```

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3? Hint: An example boxplot is provided overleaf – yours does not need to be as polished as this one.

From the box plot, it is clear that the expression of ORMDL3 is highest in the A/A genotype, intermediate in the A/G genotype, and lowest in the G/G genotype. This indicates that the G allele is linked with low expression of the ORMDL3 gene. Hence, it can be concluded that the SNP rs8067378 has an effect on the expression of the ORMDL3 gene, and the lowest expression is found in the G/G genotype.

```
boxplot(exp ~ geno,
        data = data,
        xlab = "Genotype",
        ylab = "ORMDL3 Expression",
        main = "ORMDL3 Expression by rs8067378 Genotype")

stripchart(exp ~ geno,
           data = data,
           vertical = TRUE,
           method = "jitter",
           pch = 20,
           col = "gray",
           add = TRUE)
```

### ORMDL3 Expression by rs8067378 Genotype

